Structural certificates and a correction for the A_{11} atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and structural registry,* 787 *of the* 788 *structural labels in the retained companion table headed* A_{11} *hold for their corresponding embedded subgroups. Row* 186 *prints* $2 \cdot A_5 : 2$*. Under the article's convention of left association, this expression is impossible in* A_{11}*; the corresponding subgroup instead has structure* $(A_5 \times 2) \cdot 2$*, equivalently* $2 \cdot (A_5 : 2)$ *with the displayed parentheses. An extension expression asserts a normal subgroup, quotient and the stated splitting behaviour; it need not specify a unique isomorphism class.*

Source and acquired results. The companion table by Thomas Connor and Dimitri Leemans is dated 27 November 2012 and accompanies [1]. Its numerical catalogue, source correspondence, embedded generators and orders are acquired from the separately accepted computation in degree 11. There are 788 conjugacy classes and 3304 weighted maximal-subgroup edges. The subgroup census and incidence counts are not repeated here. The registry contains 638 finite models reconstructed from previously accepted applications; the files used to reconstruct them are bound to those reviews. The acquired hypotheses and qualifications remain part of the present statement.

Section 3.2 of the article explicitly gives left association for unparenthesized extensions and distinguishes split from nonsplit extensions. We acquire that convention from its separately reviewed native reading. A new source audit joins the native powers, subscripts, parentheses and operators of all 788 labels to the retained expressions. The table has 41 pages and 17498 retained text states. Native pages $1, 2, 11, 13, 18, 40, 41$ were inspected directly; the remaining pages have position checks, not a claim of visual inspection. The position audit uses the same pypdf decoder as the flat extraction. Row 357 places redundant parentheses around the single atom D_{10}; only that exact glyph sequence is normalized. The final caption is checked explicitly and excluded from the last cell. Every printed expression, including row 186, is retained unchanged. The claim concerns this pinned edition of the table.

Lemma 1. *Let* H *be generated by specified permutations and let* S *be the union of their supports, with* $n = |S|$*. If the order of this same embedded group is* $n!$*, then* $H = \mathrm{Sym}(S)$*. If all its specified generators are even and its order is* $n!/2$*, then* $H = \mathrm{Alt}(S)$*.*

Proof. Every generator fixes the complement of S, so $H \leq \mathrm{Sym}(S)$. Equality of finite orders proves the first assertion. For the second, the sign homomorphism gives $H \leq \mathrm{Alt}(S)$, and equality of orders again gives equality of groups. The application first joins each source row to the accepted numerical record and exact generator tuple. The order used here is consequently the order of that group, rather than an inference from its printed name. □

This identifies rows $1, 2, 4, 12, 39, 138, 414, 706$. Symmetric subgroups embedded in an alternating group can have extra moved points, so a second criterion is useful for eight further rows.

Lemma 2. *Let* H *act on a finite set* X*, let* S *be invariant, and put* $n = |S| \geq 3$*. Suppose that* t *is an involution on* $X \setminus S$*, possibly the identity, and that each specified generator* h *acts on the complement as* $t^{\epsilon(h|_S)}$*, where* ϵ *is the additive sign in* $\mathbb{Z}/2$*. If the acquired order of* H *is* $n!$*, restriction identifies* H *with* $\mathrm{Sym}(S)$*.*

Proof. The sign map is a homomorphism, so the displayed relation on generators holds on every word and its inverse. Any element acting trivially on S has sign zero and therefore fixes the complement as well. Thus restriction is injective. Its image in $\mathrm{Sym}(S)$ has order $n!$, so it is the whole symmetric group. The implementation finds actual generator orbits, computes each restriction sign by inversion parity, and checks the same complement involution on every generator. It does not enumerate H or infer its order from this test. □

The second lemma applies to rows $3, 9, 26, 82, 277, 287, 762, 766$. It is the usual argument for the graph of a homomorphism. The adaptation here recognizes the sign action on the complementary points.

Lemma 3. *Let M and H be finite groups given by complete closures of specified permutation generators. A bijection $f : M \to H$ with $f(1) = 1$ and $f(xg) = f(x)f(g)$ for every $x \in M$ and every specified generator g is an isomorphism.*

Proof. Every element is a positive word in the generators: in a finite group the inverse of a generator is a nonnegative power of it. Induction on such a word proves $f(xy) = f(x)f(y)$. The reader constructs the proposed images throughout the complete closure, checks consistency on repeated elements and checks every right-generator edge. It verifies that the image is exactly the target subgroup. Thus the checked map is a bijective homomorphism, rather than only a correspondence between generating sets. □

Products and extensions. The new registry has 155 models: 96 split extensions, 30 direct products, 28 nonsplit extensions and one dicyclic group. Each model is proved before any later model or source occurrence can use it. For an extension model H, the reader closes the displayed subgroup N, checks $N \triangleleft H$, and transports an earlier proved model onto N. A split certificate supplies a subgroup K of the indicated type, checks $N \cap K = 1$ and $|N||K| = |H|$, and hence proves $H = NK$. For the colon notation it also witnesses nontrivial conjugation of K on N. A direct product instead checks commutation for all pairs of factor generators; the factors then commute elementwise.

Fifteen nonsplit models have quotient C_2. A complement exists exactly when there is an involution outside N. Indeed, such an element generates a subgroup disjoint from N whose product with N has order $|H|$. The reader checks every outside element, so absence of an involution proves nonsplitting.

The other thirteen nonsplit models have quotient S_3. The quotient certificate identifies generators with defining relations $a^2 = b^3 = (ab)^2 = 1$. A complement provides lifts satisfying the same relations. Conversely, such lifts define a homomorphism from S_3 whose composition with the quotient map is the identity, and hence a section. All possible lifts are considered. Those failing the first two relations may be rejected before the third is checked; the recorded pair count still includes the full original Cartesian product. Every pair fails at least one relation, which excludes a section.

For the source symbol Q_{28}, the new literal model has generators x, y of a group of order 28, with

$$|x| = 14, \qquad y^2 = x^7, \qquad y^{-1}xy = x^{-1}.$$

These relations reduce every word to x^i or $x^i y$, with $0 \le i < 14$. Thus the presented dicyclic group has at most 28 elements. The actual generated permutation group has exactly 28, so its canonical quotient map is an isomorphism. The inherited predicate already checked these relations, complete generation and cardinality. Its finite allowed list was extended from orders $8, 12, 16, 20$ to include 28; no relation was weakened. The earlier prepared requests were not executed.

The separate correction. The accepted companion note [2] proves that a nonsplit extension $2 \cdot A_5$ cannot embed in A_{11}. Such an extension is perfect and has a central involution. Its restriction to the support of that involution would lie in $C_2 \wr S_m$ for $m = 2$ or 4, a solvable group. A perfect group has trivial image there, contradicting the involution's nontrivial restriction. The note also identifies the actual row-186 subgroup as the parity pullback

$$\{(\sigma, k) \in S_5 \times C_4 : \operatorname{sgn}(\sigma) = (-1)^k\}.$$

Its subgroup of index two is $A_5 \times C_2$, and every element outside it has nontrivial square in the C_4 coordinate. The extension is therefore nonsplit. The full proof and its literal 240-element certificate are acquired from the distinct review of that note. The present reader replays the certificate on the exact saved generators. The correction remains separate from the 787 recovered printed labels; it is not an open-conjecture resolution.

Verification and attribution. The complete literal check passed in 23.697 seconds, with 788 source joins, 155 new models, 1051 transports and 3892103 generator edges. It performed 19706 outside-element tests and considered 87108 quotient lift pairs. The support and restriction lemmas each account for eight rows; 771 source occurrences use literal transports, and row 186 uses its separate corrected certificate. Inputs remained unchanged and stderr was empty. The native source wrapper independently replayed all 17498 retained text states and all 788 glyph joins.

Fourteen damaged source inputs are rejected, including changes to the printed row-186 assertion and the final caption. Five narrow parenthesis controls permit only the single D_{10} normalization. The two short recognition lemmas have their separate literal controls. Eleven damaged Q_{28} certificates are rejected in ordinary and optimized Python, including a cyclic group of order 28 satisfying the order and square conditions but failing the inversion relation.

The numerical catalogue, source expressions and notation belong to Connor and Leemans. The finite presentation, restriction, transport and extension arguments are standard group theory or acquired common results. The work here provides the explicit joins and finite certificates for this table, together with the separately proved correction. Candidate searches supply witnesses only; the literal reader establishes their assertions. Distinct final source admission remains to be recorded. No Lean formalization, whole-article completion or compression measurement is claimed here. Other large companion tables remain in scope for L054.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed A_{11} and dated 27 November 2012.

[2] human_compression project, *A structural label in the subgroup table of* A_{11}, companion proof and literal certificate, acquired in the present packet.